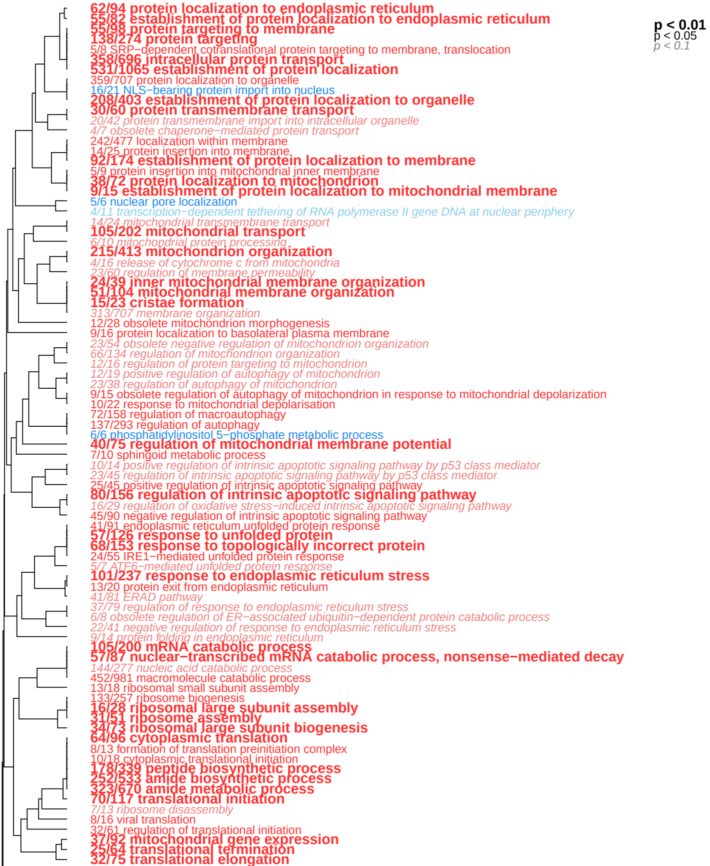
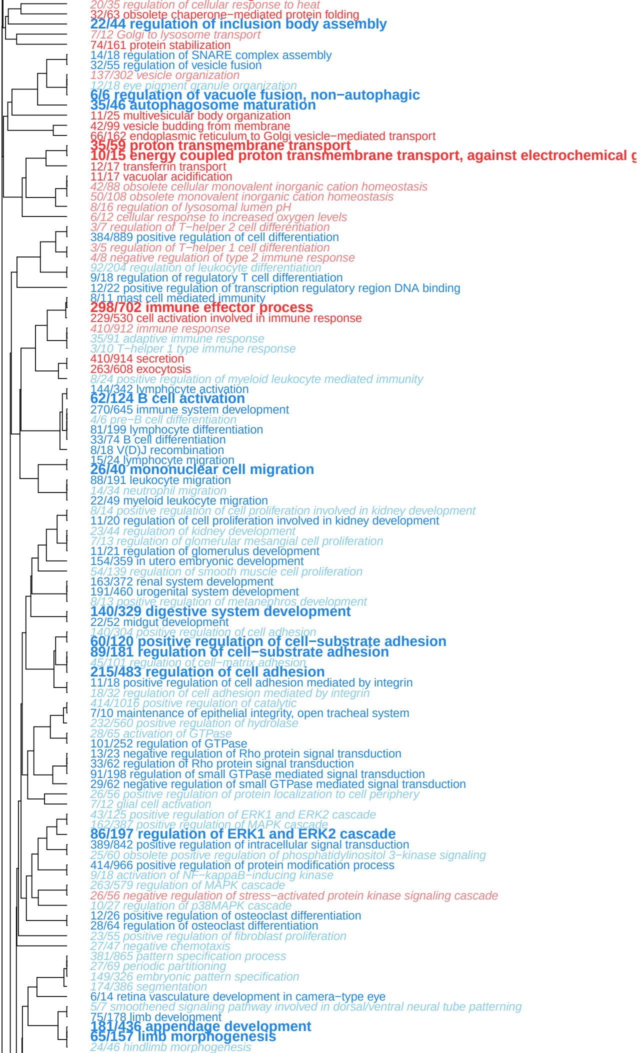
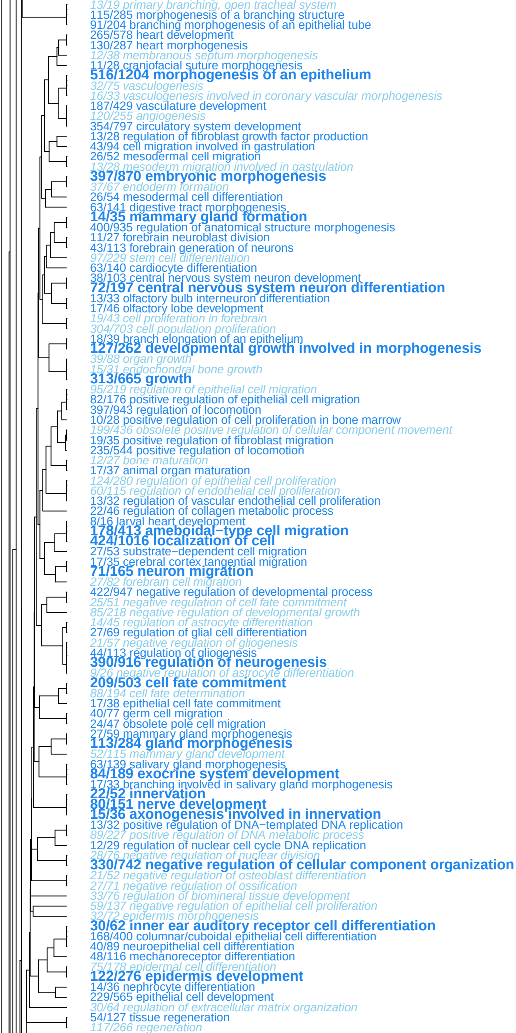
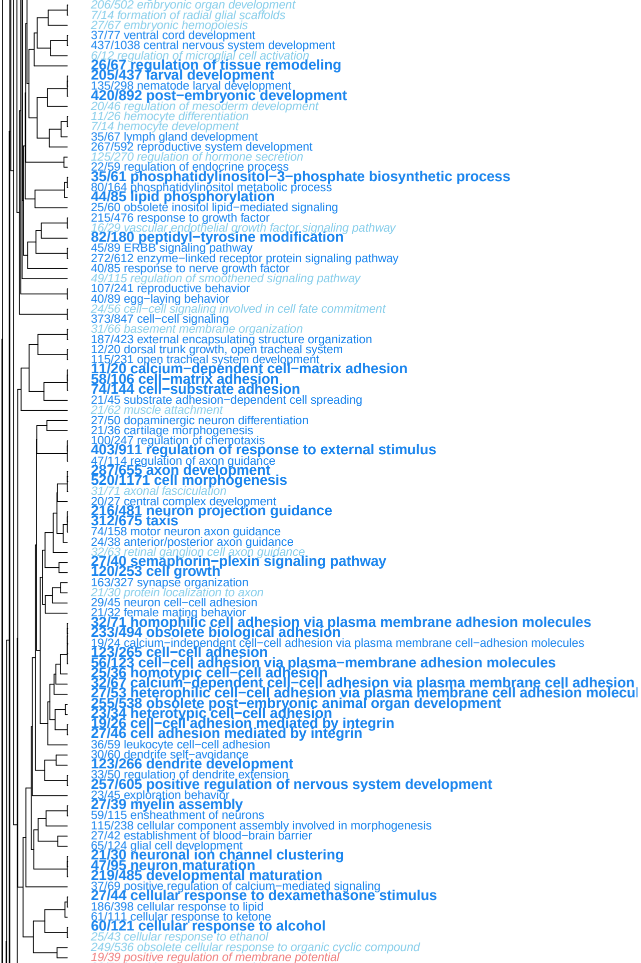
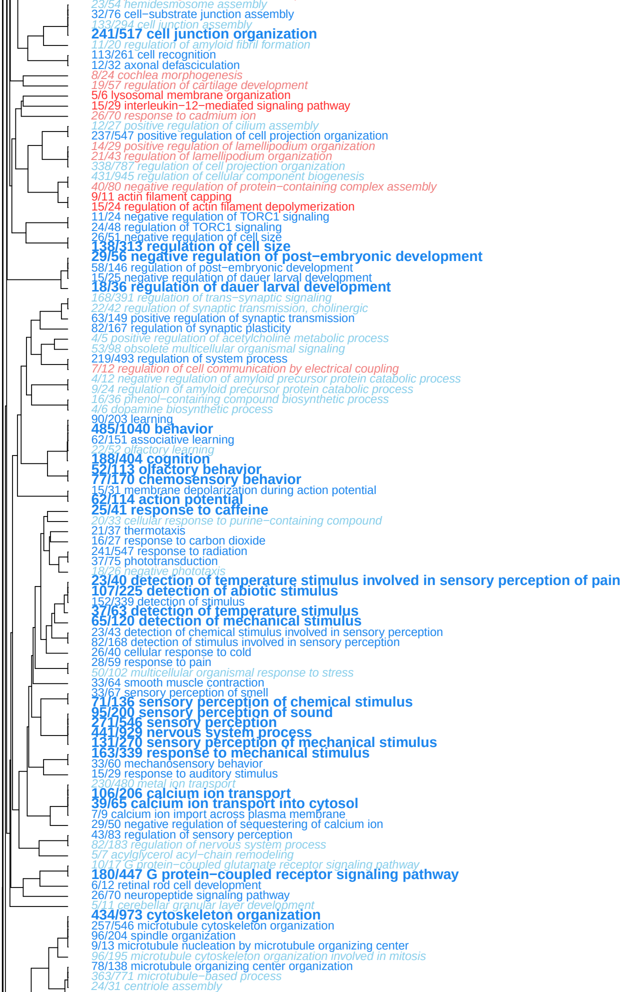
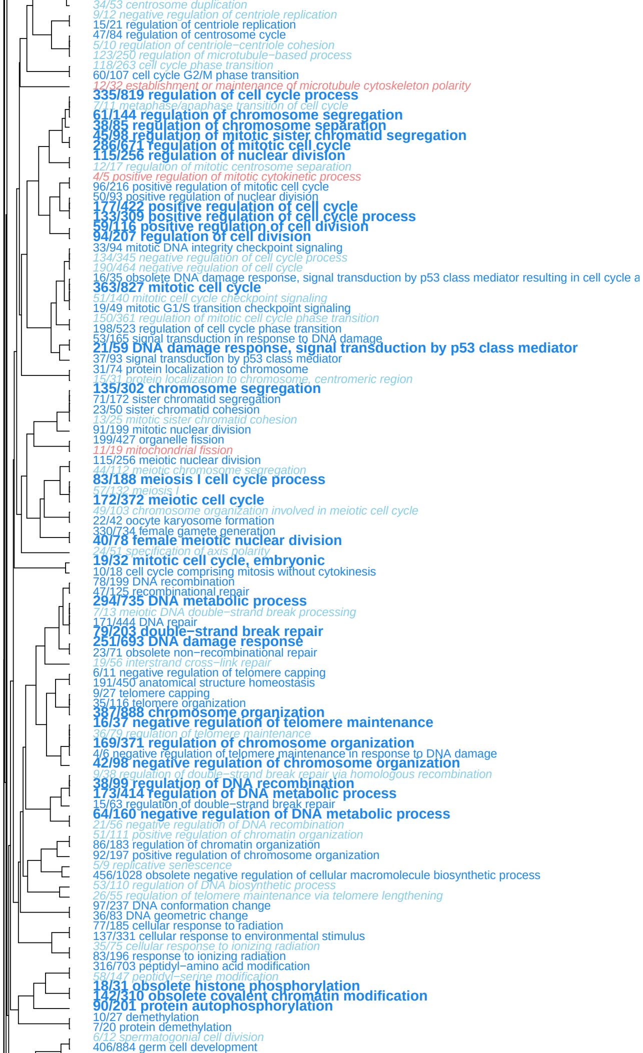

62/94 protein localization to endoplasmic reticulum
55/82 establishment of protein localization to endoplasmic reticulum
55/98 protein targeting to membrane
138/274 protein targeting
5/8 SRP−dependent cotranslational protein targeting to membrane, translocation
358/696 intracellular protein transport
531/1065 establishment of protein localization
359/707 protein localization to organelle
16/21 NLS−bearing protein import into nucleus
208/403 establishment of protein localization to organelle
30/60 protein transmembrane transport
20/42 protein transmembrane import into intracellular organelle
4/7 obsolete chaperone−mediated protein transport
242/477 localization within membrane
14/25 protein insertion into membrane
92/174 establishment of protein localization to membrane
5/9 protein insertion into mitochondrial inner membrane
38/72 protein localization to mitochondrion
9/15 establishment of protein localization to mitochondrial membrane
5/6 nuclear pore localization
4/11 transcription−dependent tethering of RNA polymerase II gene DNA at nuclear periphery
14/24 mitochondrial transmembrane transport
105/202 mitochondrial transport
6/10 mitochondrial protein processing
215/413 mitochondrion organization
4/16 release of cytochrome c from mitochondria
23/60 regulation of membrane permeability
24/39 inner mitochondrial membrane organization
51/104 mitochondrial membrane organization
15/23 cristae formation
313/707 membrane organization
12/28 obsolete mitochondrion morphogenesis
9/16 protein localization to basolateral plasma membrane
23/54 obsolete negative regulation of mitochondrion organization
66/134 regulation of mitochondrion organization
12/16 regulation of protein targeting to mitochondrion
12/19 positive regulation of autophagy of mitochondrion
23/38 regulation of autophagy of mitochondrion
9/15 obsolete regulation of autophagy of mitochondrion in response to mitochondrial depolarization
10/22 response to mitochondrial depolarisation
72/158 regulation of macroautophagy
137/293 regulation of autophagy
6/6 phosphatidylinositol 5−phosphate metabolic process
40/75 regulation of mitochondrial membrane potential
7/10 sphingoid metabolic process
10/14 positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator
23/45 regulation of intrinsic apoptotic signaling pathway by p53 class mediator
25/45 positive regulation of intrinsic apoptotic signaling pathway
80/156 regulation of intrinsic apoptotic signaling pathway
16/29 regulation of oxidative stress−induced intrinsic apoptotic signaling pathway
45/90 negative regulation of intrinsic apoptotic signaling pathway
41/91 endoplasmic reticulum unfolded protein response
57/126 response to unfolded protein
68/153 response to topologically incorrect protein
24/55 IRE1−mediated unfolded protein response
5/7 ATF6−mediated unfolded protein response
101/237 response to endoplasmic reticulum stress
13/20 protein exit from endoplasmic reticulum
41/81 ERAD pathway
37/79 regulation of response to endoplasmic reticulum stress
6/8 obsolete regulation of ER−associated ubiquitin−dependent protein catabolic process
22/41 negative regulation of response to endoplasmic reticulum stress
9/14 protein folding in endoplasmic reticulum
105/200 mRNA catabolic process
57/87 nuclear−transcribed mRNA catabolic process, nonsense−mediated decay
144/277 nucleic acid catabolic process
452/981 macromolecule catabolic process
13/18 ribosomal small subunit assembly
133/257 ribosome biogenesis
16/28 ribosomal large subunit assembly
31/51 ribosome assembly
34/73 ribosomal large subunit biogenesis
64/96 cytoplasmic translation
8/13 formation of translation preinitiation complex
10/18 cytoplasmic translational initiation
178/339 peptide biosynthetic process
252/533 amide biosynthetic process
323/670 amide metabolic process
70/117 translational initiation
7/13 ribosome disassembly
8/16 viral translation
32/61 regulation of translational initiation
37/92 mitochondrial gene expression
25/64 translational termination
32/75 translational elongation
p < 0.01
p < 0.05
p < 0.1

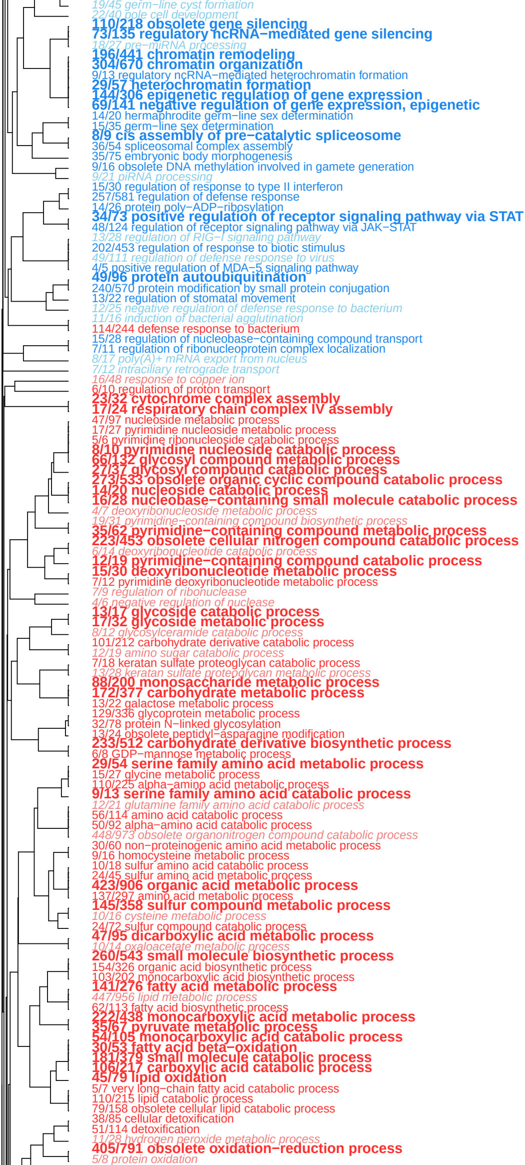

20/35 regulation of cellular response to heat
32/63 obsolete chaperone−mediated protein folding
22/44 regulation of inclusion body assembly
7/12 Golgi to lysosome transport
74/161 protein stabilization
14/18 regulation of SNARE complex assembly
32/55 regulation of vesicle fusion
137/302 vesicle organization
12/18 eye pigment granule organization
6/6 regulation of vacuole fusion, non−autophagic
35/46 autophagosome maturation
11/25 multivesicular body organization
42/99 vesicle budding from membrane
66/162 endoplasmic reticulum to Golgi vesicle−mediated transport
35/59 proton transmembrane transport
10/15 energy coupled proton transmembrane transport, against electrochemical g
12/17 transferrin transport
11/17 vacuolar acidification
42/88 obsolete cellular monovalent inorganic cation homeostasis
50/108 obsolete monovalent inorganic cation homeostasis
8/16 regulation of lysosomal lumen pH
6/12 cellular response to increased oxygen levels
3/7 regulation of T−helper 2 cell differentiation
384/889 positive regulation of cell differentiation
3/5 regulation of T−helper 1 cell differentiation
4/8 negative regulation of type 2 immune response
92/204 regulation of leukocyte differentiation
9/18 regulation of regulatory T cell differentiation
12/22 positive regulation of transcription regulatory region DNA binding
8/11 mast cell mediated immunity
298/702 immune effector process
229/530 cell activation involved in immune response
410/912 immune response
35/91 adaptive immune response
3/10 T−helper 1 type immune response
410/914 secretion
263/608 exocytosis
8/24 positive regulation of myeloid leukocyte mediated immunity
144/342 lymphocyte activation
62/124 B cell activation
270/645 immune system development
4/6 pre−B cell differentiation
81/199 lymphocyte differentiation
33/74 B cell differentiation
8/18 V(D)J recombination
15/24 lymphocyte migration
26/40 mononuclear cell migration
88/191 leukocyte migration
14/34 neutrophil migration
22/49 myeloid leukocyte migration
8/14 positive regulation of cell proliferation involved in kidney development
11/20 regulation of cell proliferation involved in kidney development
23/44 regulation of kidney development
7/13 regulation of glomerular mesangial cell proliferation
11/21 regulation of glomerulus development
154/359 in utero embryonic development
54/139 regulation of smooth muscle cell proliferation
163/372 renal system development
191/460 urogenital system development
8/13 positive regulation of metanephros development
140/329 digestive system development
22/52 midgut development
140/304 positive regulation of cell adhesion
60/120 positive regulation of cell−substrate adhesion
89/181 regulation of cell−substrate adhesion
45/101 regulation of cell−matrix adhesion
215/483 regulation of cell adhesion
11/18 positive regulation of cell adhesion mediated by integrin
18/32 regulation of cell adhesion mediated by integrin
414/1016 positive regulation of catalytic
7/10 maintenance of epithelial integrity, open tracheal system
232/560 positive regulation of hydrolase
28/65 activation of GTPase
101/252 regulation of GTPase
13/23 negative regulation of Rho protein signal transduction
33/62 regulation of Rho protein signal transduction
91/198 regulation of small GTPase mediated signal transduction
29/62 negative regulation of small GTPase mediated signal transduction
26/56 positive regulation of protein localization to cell periphery
7/12 glial cell activation
43/125 positive regulation of ERK1 and ERK2 cascade
162/387 positive regulation of MAPK cascade
86/197 regulation of ERK1 and ERK2 cascade
389/842 positive regulation of intracellular signal transduction
25/60 obsolete positive regulation of phosphatidylinositol 3−kinase signaling
414/966 positive regulation of protein modification process
9/18 activation of NF−kappaB−inducing kinase
263/579 regulation of MAPK cascade
26/56 negative regulation of stress−activated protein kinase signaling cascade
10/27 regulation of p38MAPK cascade
12/26 positive regulation of osteoclast differentiation
28/64 regulation of osteoclast differentiation
23/55 positive regulation of fibroblast proliferation
27/47 negative chemotaxis
381/865 pattern specification process
27/69 periodic partitioning
149/326 embryonic pattern specification
174/386 segmentation
6/14 retina vasculature development in camera−type eye
5/7 smoothened signaling pathway involved in dorsal/ventral neural tube patterning
75/178 limb development
181/436 appendage development
65/157 limb morphogenesis
24/46 hindlimb morphogenesis

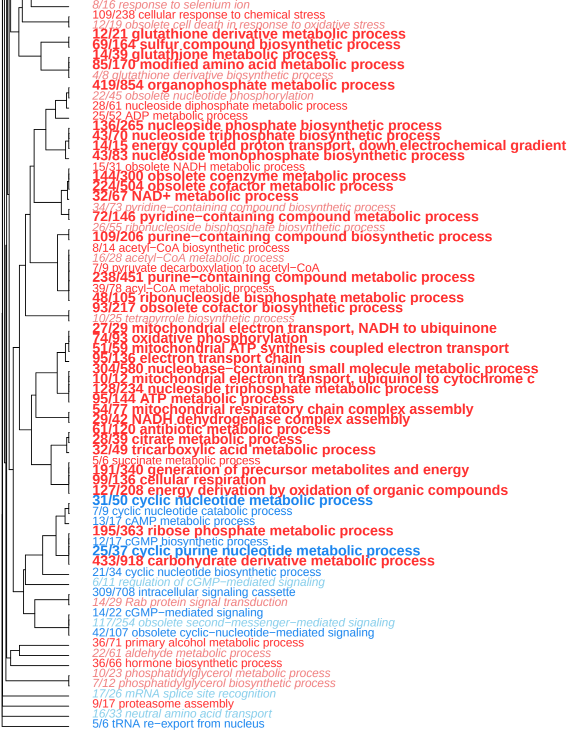

13/19 primary branching, open tracheal system
115/285 morphogenesis of a branching structure
91/204 branching morphogenesis of an epithelial tube
265/578 heart development
130/287 heart morphogenesis
12/38 membranous septum morphogenesis
11/28 craniofacial suture morphogenesis
516/1204 morphogenesis of an epithelium
32/75 vasculogenesis
16/33 vasculogenesis involved in coronary vascular morphogenesis
187/429 vasculature development
120/255 angiogenesis
354/797 circulatory system development
13/28 regulation of fibroblast growth factor production
43/94 cell migration involved in gastrulation
26/52 mesodermal cell migration
13/28 mesoderm migration involved in gastrulation
397/870 embryonic morphogenesis
37/67 endoderm formation
26/54 mesodermal cell differentiation
63/141 digestive tract morphogenesis
14/35 mammary gland formation
400/935 regulation of anatomical structure morphogenesis
11/27 forebrain neuroblast division
43/113 forebrain generation of neurons
97/229 stem cell differentiation
63/140 cardiocyte differentiation
38/103 central nervous system neuron development
72/197 central nervous system neuron differentiation
13/33 olfactory bulb interneuron differentiation
17/46 olfactory lobe development
19/43 cell proliferation in forebrain
304/703 cell population proliferation
18/39 branch elongation of an epithelium
127/262 developmental growth involved in morphogenesis
39/88 organ growth
15/31 endochondral bone growth
313/665 growth
95/219 regulation of epithelial cell migration
82/176 positive regulation of epithelial cell migration
397/943 regulation of locomotion
10/28 positive regulation of cell proliferation in bone marrow
199/436 obsolete positive regulation of cellular component movement
19/35 positive regulation of fibroblast migration
235/544 positive regulation of locomotion
12/27 bone maturation
17/37 animal organ maturation
124/280 regulation of epithelial cell proliferation
60/115 regulation of endothelial cell proliferation
13/32 regulation of vascular endothelial cell proliferation
22/46 regulation of collagen metabolic process
8/16 larval heart development
178/413 ameboidal−type cell migration
424/1016 localization of cell
27/53 substrate−dependent cell migration
17/35 cerebral cortex tangential migration
71/165 neuron migration
27/82 forebrain cell migration
422/947 negative regulation of developmental process
25/51 negative regulation of cell fate commitment
85/218 negative regulation of developmental growth
14/45 regulation of astrocyte differentiation
27/69 regulation of glial cell differentiation
21/57 negative regulation of gliogenesis
44/113 regulation of gliogenesis
390/916 regulation of neurogenesis
9/26 negative regulation of astrocyte differentiation
209/503 cell fate commitment
88/194 cell fate determination
17/38 epithelial cell fate commitment
40/77 germ cell migration
24/47 obsolete pole cell migration
27/59 mammary gland morphogenesis
113/284 gland morphogenesis
52/115 mammary gland development
63/139 salivary gland morphogenesis
84/189 exocrine system development
17/33 branching involved in salivary gland morphogenesis
22/52 innervation
80/151 nerve development
15/36 axonogenesis involved in innervation
13/32 positive regulation of DNA−templated DNA replication
89/227 positive regulation of DNA metabolic process
12/29 regulation of nuclear cell cycle DNA replication
28/76 negative regulation of nuclear division
330/742 negative regulation of cellular component organization
21/52 negative regulation of osteoblast differentiation
27/71 negative regulation of ossification
33/76 regulation of biomineral tissue development
59/137 negative regulation of epithelial cell proliferation
32/72 epidermis morphogenesis
30/62 inner ear auditory receptor cell differentiation
168/400 columnar/cuboidal epithelial cell differentiation
40/89 neuroepithelial cell differentiation
48/116 mechanoreceptor differentiation
75/178 epidermal cell differentiation
122/276 epidermis development
14/36 nephrocyte differentiation
229/565 epithelial cell development
30/64 regulation of extracellular matrix organization
54/127 tissue regeneration
117/266 regeneration


206/502 embryonic organ development
7/14 formation of radial glial scaffolds
27/67 embryonic hemopoiesis
37/77 ventral cord development
437/1038 central nervous system development
6/12 regulation of microglial cell activation
26/67 regulation of tissue remodeling
205/437 larval development
135/298 nematode larval development
420/892 post−embryonic development
20/46 regulation of mesoderm development
11/26 hemocyte differentiation
7/14 hemocyte development
35/67 lymph gland development
267/592 reproductive system development
125/270 regulation of hormone secretion
22/59 regulation of endocrine process
35/61 phosphatidylinositol−3−phosphate biosynthetic process
80/164 phosphatidylinositol metabolic process
44/85 lipid phosphorylation
25/60 obsolete inositol lipid−mediated signaling
215/476 response to growth factor
16/29 vascular endothelial growth factor signaling pathway
82/180 peptidyl−tyrosine modification
45/89 ERBB signaling pathway
272/612 enzyme−linked receptor protein signaling pathway
40/85 response to nerve growth factor
49/115 regulation of smoothened signaling pathway
107/241 reproductive behavior
40/89 egg−laying behavior
24/56 cell−cell signaling involved in cell fate commitment
373/847 cell−cell signaling
31/66 basement membrane organization
187/423 external encapsulating structure organization
12/20 dorsal trunk growth, open tracheal system
115/231 open tracheal system development
11/20 calcium−dependent cell−matrix adhesion
58/106 cell−matrix adhesion
74/144 cell−substrate adhesion
21/45 substrate adhesion−dependent cell spreading
21/62 muscle attachment
27/50 dopaminergic neuron differentiation
21/36 cartilage morphogenesis
100/247 regulation of chemotaxis
403/911 regulation of response to external stimulus
47/114 regulation of axon guidance
287/655 axon development
520/1171 cell morphogenesis
31/71 axonal fasciculation
20/27 central complex development
216/481 neuron projection guidance
312/675 taxis
74/158 motor neuron axon guidance
24/38 anterior/posterior axon guidance
32/63 retinal ganglion cell axon guidance
27/40 semaphorin−plexin signaling pathway
120/253 cell growth
163/327 synapse organization
21/30 protein localization to axon
29/45 neuron cell−cell adhesion
21/32 female mating behavior
32/71 homophilic cell adhesion via plasma membrane adhesion molecules
233/494 obsolete biological adhesion
19/24 calcium−independent cell−cell adhesion via plasma membrane cell−adhesion molecules
123/265 cell−cell adhesion
56/123 cell−cell adhesion via plasma−membrane adhesion molecules
25/36 homotypic cell−cell adhesion
32/67 calcium−dependent cell−cell adhesion via plasma membrane cell adhesion
27/53 heterophilic cell−cell adhesion via plasma membrane cell adhesion molecul
255/538 obsolete post−embryonic animal organ development
23/34 heterotypic cell−cell adhesion
19/26 cell−cell adhesion mediated by integrin
27/46 cell adhesion mediated by integrin
36/59 leukocyte cell−cell adhesion
30/60 dendrite self−avoidance
123/266 dendrite development
33/50 regulation of dendrite extension
257/605 positive regulation of nervous system development
23/45 exploration behavior
27/39 myelin assembly
59/115 ensheathment of neurons
115/238 cellular component assembly involved in morphogenesis
27/42 establishment of blood−brain barrier
65/124 glial cell development
21/30 neuronal ion channel clustering
47/95 neuron maturation
219/485 developmental maturation
37/69 positive regulation of calcium−mediated signaling
27/44 cellular response to dexamethasone stimulus
186/398 cellular response to lipid
61/111 cellular response to ketone
60/121 cellular response to alcohol
25/43 cellular response to ethanol
249/536 obsolete cellular response to organic cyclic compound
19/39 positive regulation of membrane potential


23/54 hemidesmosome assembly
32/76 cell−substrate junction assembly
133/294 cell junction assembly
241/517 cell junction organization
11/20 regulation of amyloid fibril formation
113/261 cell recognition
12/32 axonal defasciculation
8/24 cochlea morphogenesis
19/57 regulation of cartilage development
5/6 lysosomal membrane organization
15/29 interleukin−12−mediated signaling pathway
26/70 response to cadmium ion
12/27 positive regulation of cilium assembly
237/547 positive regulation of cell projection organization
14/29 positive regulation of lamellipodium organization
21/43 regulation of lamellipodium organization
338/787 regulation of cell projection organization
431/945 regulation of cellular component biogenesis
40/80 negative regulation of protein−containing complex assembly
9/11 actin filament capping
15/24 regulation of actin filament depolymerization
11/24 negative regulation of TORC1 signaling
24/48 regulation of TORC1 signaling
26/51 negative regulation of cell size
138/313 regulation of cell size
29/56 negative regulation of post−embryonic development
58/146 regulation of post−embryonic development
15/25 negative regulation of dauer larval development
18/36 regulation of dauer larval development
168/391 regulation of trans−synaptic signaling
22/42 regulation of synaptic transmission, cholinergic
63/149 positive regulation of synaptic transmission
82/167 regulation of synaptic plasticity
4/5 positive regulation of acetylcholine metabolic process
53/98 obsolete multicellular organismal signaling
219/493 regulation of system process
7/12 regulation of cell communication by electrical coupling
4/12 negative regulation of amyloid precursor protein catabolic process
9/24 regulation of amyloid precursor protein catabolic process
16/36 phenol−containing compound biosynthetic process
4/6 dopamine biosynthetic process
90/203 learning
485/1040 behavior
62/151 associative learning
22/52 olfactory learning
188/404 cognition
52/113 olfactory behavior
77/170 chemosensory behavior
15/31 membrane depolarization during action potential
62/114 action potential
25/41 response to caffeine
20/33 cellular response to purine−containing compound
21/37 thermotaxis
16/27 response to carbon dioxide
241/547 response to radiation
37/75 phototransduction
18/26 negative phototaxis
23/40 detection of temperature stimulus involved in sensory perception of pain
107/225 detection of abiotic stimulus
152/339 detection of stimulus
37/63 detection of temperature stimulus
65/120 detection of mechanical stimulus
23/43 detection of chemical stimulus involved in sensory perception
82/168 detection of stimulus involved in sensory perception
26/40 cellular response to cold
28/59 response to pain
50/102 multicellular organismal response to stress
33/64 smooth muscle contraction
33/67 sensory perception of smell
71/136 sensory perception of chemical stimulus
95/200 sensory perception of sound
271/546 sensory perception
441/929 nervous system process
131/270 sensory perception of mechanical stimulus
163/339 response to mechanical stimulus
33/60 mechanosensory behavior
15/29 response to auditory stimulus
230/480 metal ion transport
106/206 calcium ion transport
39/65 calcium ion transport into cytosol
7/9 calcium ion import across plasma membrane
29/50 negative regulation of sequestering of calcium ion
43/83 regulation of sensory perception
82/183 regulation of nervous system process
5/7 acylglycerol acyl−chain remodeling
10/17 G protein−coupled glutamate receptor signaling pathway
180/447 G protein−coupled receptor signaling pathway
6/12 retinal rod cell development
26/70 neuropeptide signaling pathway
5/11 cerebellar granular layer development
434/973 cytoskeleton organization
257/546 microtubule cytoskeleton organization
96/204 spindle organization
9/13 microtubule nucleation by microtubule organizing center
96/195 microtubule cytoskeleton organization involved in mitosis
78/138 microtubule organizing center organization
363/771 microtubule−based process
24/31 centriole assembly


34/53 centrosome duplication
9/12 negative regulation of centriole replication
15/21 regulation of centriole replication
47/84 regulation of centrosome cycle
5/10 regulation of centriole−centriole cohesion
123/250 regulation of microtubule−based process
118/263 cell cycle phase transition
60/107 cell cycle G2/M phase transition
12/32 establishment or maintenance of microtubule cytoskeleton polarity
335/819 regulation of cell cycle process
7/11 metaphase/anaphase transition of cell cycle
61/144 regulation of chromosome segregation
38/85 regulation of chromosome separation
45/98 regulation of mitotic sister chromatid segregation
286/671 regulation of mitotic cell cycle
115/256 regulation of nuclear division
12/17 regulation of mitotic centrosome separation
4/5 positive regulation of mitotic cytokinetic process
96/216 positive regulation of mitotic cell cycle
50/93 positive regulation of nuclear division
177/422 positive regulation of cell cycle
133/309 positive regulation of cell cycle process
59/116 positive regulation of cell division
94/207 regulation of cell division
33/94 mitotic DNA integrity checkpoint signaling
134/345 negative regulation of cell cycle process
190/464 negative regulation of cell cycle
16/35 obsolete DNA damage response, signal transduction by p53 class mediator resulting in cell cycle a
363/827 mitotic cell cycle
51/140 mitotic cell cycle checkpoint signaling
19/49 mitotic G1/S transition checkpoint signaling
150/361 regulation of mitotic cell cycle phase transition
198/523 regulation of cell cycle phase transition
53/165 signal transduction in response to DNA damage
21/59 DNA damage response, signal transduction by p53 class mediator
37/93 signal transduction by p53 class mediator
31/74 protein localization to chromosome
15/31 protein localization to chromosome, centromeric region
135/302 chromosome segregation
71/172 sister chromatid segregation
23/50 sister chromatid cohesion
13/25 mitotic sister chromatid cohesion
91/199 mitotic nuclear division
199/427 organelle fission
11/19 mitochondrial fission
115/256 meiotic nuclear division
44/112 meiotic chromosome segregation
83/188 meiosis I cell cycle process
57/132 meiosis I
172/372 meiotic cell cycle
49/103 chromosome organization involved in meiotic cell cycle
22/42 oocyte karyosome formation
330/734 female gamete generation
40/78 female meiotic nuclear division
24/51 specification of axis polarity
19/32 mitotic cell cycle, embryonic
10/18 cell cycle comprising mitosis without cytokinesis
78/199 DNA recombination
47/125 recombinational repair
294/735 DNA metabolic process
7/13 meiotic DNA double−strand break processing
171/444 DNA repair
79/203 double−strand break repair
251/693 DNA damage response
23/71 obsolete non−recombinational repair
19/56 interstrand cross−link repair
6/11 negative regulation of telomere capping
191/450 anatomical structure homeostasis
9/27 telomere capping
35/116 telomere organization
387/888 chromosome organization
16/37 negative regulation of telomere maintenance
36/79 regulation of telomere maintenance
169/371 regulation of chromosome organization
4/6 negative regulation of telomere maintenance in response to DNA damage
42/98 negative regulation of chromosome organization
9/38 regulation of double−strand break repair via homologous recombination
38/99 regulation of DNA recombination
173/414 regulation of DNA metabolic process
15/63 regulation of double−strand break repair
64/160 negative regulation of DNA metabolic process
21/56 negative regulation of DNA recombination
51/111 positive regulation of chromatin organization
86/183 regulation of chromatin organization
92/197 positive regulation of chromosome organization
5/9 replicative senescence
456/1028 obsolete negative regulation of cellular macromolecule biosynthetic process
53/110 regulation of DNA biosynthetic process
26/55 regulation of telomere maintenance via telomere lengthening
97/237 DNA conformation change
36/83 DNA geometric change
77/185 cellular response to radiation
137/331 cellular response to environmental stimulus
35/75 cellular response to ionizing radiation
83/196 response to ionizing radiation
316/703 peptidyl−amino acid modification
58/147 peptidyl−serine modification
18/31 obsolete histone phosphorylation
142/310 obsolete covalent chromatin modification
90/201 protein autophosphorylation
10/27 demethylation
7/20 protein demethylation
6/12 spermatogonial cell division
406/884 germ cell development


19/45 germ−line cyst formation
22/40 pole cell development
110/218 obsolete gene silencing
73/135 regulatory ncRNA−mediated gene silencing
18/27 pre−miRNA processing
196/441 chromatin remodeling
304/670 chromatin organization
9/13 regulatory ncRNA−mediated heterochromatin formation
29/57 heterochromatin formation
144/306 epigenetic regulation of gene expression
69/141 negative regulation of gene expression, epigenetic
14/20 hermaphrodite germ−line sex determination
15/35 germ−line sex determination
8/9 cis assembly of pre−catalytic spliceosome
36/54 spliceosomal complex assembly
35/75 embryonic body morphogenesis
9/16 obsolete DNA methylation involved in gamete generation
9/21 piRNA processing
15/30 regulation of response to type II interferon
257/581 regulation of defense response
14/26 protein poly−ADP−ribosylation
34/73 positive regulation of receptor signaling pathway via STAT
48/124 regulation of receptor signaling pathway via JAK−STAT
13/28 regulation of RIG−I signaling pathway
202/453 regulation of response to biotic stimulus
49/111 regulation of defense response to virus
4/5 positive regulation of MDA−5 signaling pathway
49/96 protein autoubiquitination
240/570 protein modification by small protein conjugation
13/22 regulation of stomatal movement
12/25 negative regulation of defense response to bacterium
11/16 induction of bacterial agglutination
114/244 defense response to bacterium
15/28 regulation of nucleobase−containing compound transport
7/11 regulation of ribonucleoprotein complex localization
8/17 poly(A)+ mRNA export from nucleus
7/12 intraciliary retrograde transport
16/48 response to copper ion
6/10 regulation of proton transport
23/32 cytochrome complex assembly
17/24 respiratory chain complex IV assembly
47/97 nucleoside metabolic process
17/27 pyrimidine nucleoside metabolic process
5/6 pyrimidine ribonucleoside catabolic process
8/10 pyrimidine nucleoside catabolic process
66/132 glycosyl compound metabolic process
27/37 glycosyl compound catabolic process
273/533 obsolete organic cyclic compound catabolic process
14/20 nucleoside catabolic process
16/28 nucleobase−containing small molecule catabolic process
4/7 deoxyribonucleoside metabolic process
19/31 pyrimidine−containing compound biosynthetic process
35/62 pyrimidine−containing compound metabolic process
223/453 obsolete cellular nitrogen compound catabolic process
6/14 deoxyribonucleotide catabolic process
12/19 pyrimidine−containing compound catabolic process
15/30 deoxyribonucleotide metabolic process
7/12 pyrimidine deoxyribonucleotide metabolic process
7/9 regulation of ribonuclease
4/6 negative regulation of nuclease
13/17 glycoside catabolic process
17/32 glycoside metabolic process
8/12 glycosylceramide catabolic process
101/212 carbohydrate derivative catabolic process
12/19 amino sugar catabolic process
7/18 keratan sulfate proteoglycan catabolic process
13/28 keratan sulfate proteoglycan metabolic process
88/200 monosaccharide metabolic process
172/377 carbohydrate metabolic process
13/22 galactose metabolic process
129/336 glycoprotein metabolic process
32/78 protein N−linked glycosylation
13/24 obsolete peptidyl−asparagine modification
233/512 carbohydrate derivative biosynthetic process
6/8 GDP−mannose metabolic process
29/54 serine family amino acid metabolic process
15/27 glycine metabolic process
110/225 alpha−amino acid metabolic process
9/13 serine family amino acid catabolic process
12/21 glutamine family amino acid catabolic process
56/114 amino acid catabolic process
50/92 alpha−amino acid catabolic process
448/973 obsolete organonitrogen compound catabolic process
30/60 non−proteinogenic amino acid metabolic process
9/16 homocysteine metabolic process
10/18 sulfur amino acid catabolic process
24/45 sulfur amino acid metabolic process
423/906 organic acid metabolic process
137/297 amino acid metabolic process
145/358 sulfur compound metabolic process
10/16 cysteine metabolic process
24/72 sulfur compound catabolic process
47/95 dicarboxylic acid metabolic process
10/14 oxaloacetate metabolic process
260/543 small molecule biosynthetic process
154/326 organic acid biosynthetic process
103/202 monocarboxylic acid biosynthetic process
141/276 fatty acid metabolic process
447/956 lipid metabolic process
62/113 fatty acid biosynthetic process
222/438 monocarboxylic acid metabolic process
35/67 pyruvate metabolic process
54/105 monocarboxylic acid catabolic process
30/53 fatty acid beta−oxidation
181/379 small molecule catabolic process
106/217 carboxylic acid catabolic process
45/79 lipid oxidation
5/7 very long−chain fatty acid catabolic process
110/215 lipid catabolic process
79/158 obsolete cellular lipid catabolic process
38/85 cellular detoxification
51/114 detoxification
11/28 hydrogen peroxide metabolic process
405/791 obsolete oxidation−reduction process
5/8 protein oxidation


8/16 response to selenium ion
109/238 cellular response to chemical stress
12/19 obsolete cell death in response to oxidative stress
12/21 glutathione derivative metabolic process
69/164 sulfur compound biosynthetic process
14/39 glutathione metabolic process
85/170 modified amino acid metabolic process
4/8 glutathione derivative biosynthetic process
419/854 organophosphate metabolic process
22/45 obsolete nucleotide phosphorylation
28/61 nucleoside diphosphate metabolic process
25/52 ADP metabolic process
136/265 nucleoside phosphate biosynthetic process
43/70 nucleoside triphosphate biosynthetic process
14/15 energy coupled proton transport, down electrochemical gradient
43/83 nucleoside monophosphate biosynthetic process
15/31 obsolete NADH metabolic process
144/300 obsolete coenzyme metabolic process
224/504 obsolete cofactor metabolic process
32/67 NAD+ metabolic process
34/73 pyridine−containing compound biosynthetic process
72/146 pyridine−containing compound metabolic process
26/55 ribonucleoside bisphosphate biosynthetic process
109/206 purine−containing compound biosynthetic process
8/14 acetyl−CoA biosynthetic process
16/28 acetyl−CoA metabolic process
7/9 pyruvate decarboxylation to acetyl−CoA
238/451 purine−containing compound metabolic process
39/78 acyl−CoA metabolic process
48/105 ribonucleoside bisphosphate metabolic process
93/217 obsolete cofactor biosynthetic process
10/25 tetrapyrrole biosynthetic process
27/29 mitochondrial electron transport, NADH to ubiquinone
74/93 oxidative phosphorylation
51/59 mitochondrial ATP synthesis coupled electron transport
95/136 electron transport chain
304/580 nucleobase−containing small molecule metabolic process
10/12 mitochondrial electron transport, ubiquinol to cytochrome c
128/234 nucleoside triphosphate metabolic process
95/144 ATP metabolic process
54/77 mitochondrial respiratory chain complex assembly
29/42 NADH dehydrogenase complex assembly
61/120 antibiotic metabolic process
28/39 citrate metabolic process
32/49 tricarboxylic acid metabolic process
5/6 succinate metabolic process
191/340 generation of precursor metabolites and energy
99/136 cellular respiration
127/208 energy derivation by oxidation of organic compounds
31/50 cyclic nucleotide metabolic process
7/9 cyclic nucleotide catabolic process
13/17 cAMP metabolic process
195/363 ribose phosphate metabolic process
12/17 cGMP biosynthetic process
25/37 cyclic purine nucleotide metabolic process
433/918 carbohydrate derivative metabolic process
21/34 cyclic nucleotide biosynthetic process
6/11 regulation of cGMP−mediated signaling
309/708 intracellular signaling cassette
14/29 Rab protein signal transduction
14/22 cGMP−mediated signaling
117/254 obsolete second−messenger−mediated signaling
42/107 obsolete cyclic−nucleotide−mediated signaling
36/71 primary alcohol metabolic process
22/61 aldehyde metabolic process
36/66 hormone biosynthetic process
10/23 phosphatidylglycerol metabolic process
7/12 phosphatidylglycerol biosynthetic process
17/26 mRNA splice site recognition
9/17 proteasome assembly
16/33 neutral amino acid transport
5/6 tRNA re−export from nucleus